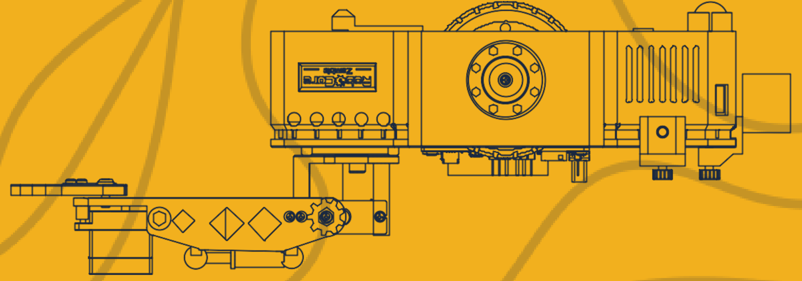
RoboCore V2 box, outside, tile 1 of 6 (row 1, column 1), 1:1. Overlap 10 mm with the neighbouring tiles.



ROBOCORE

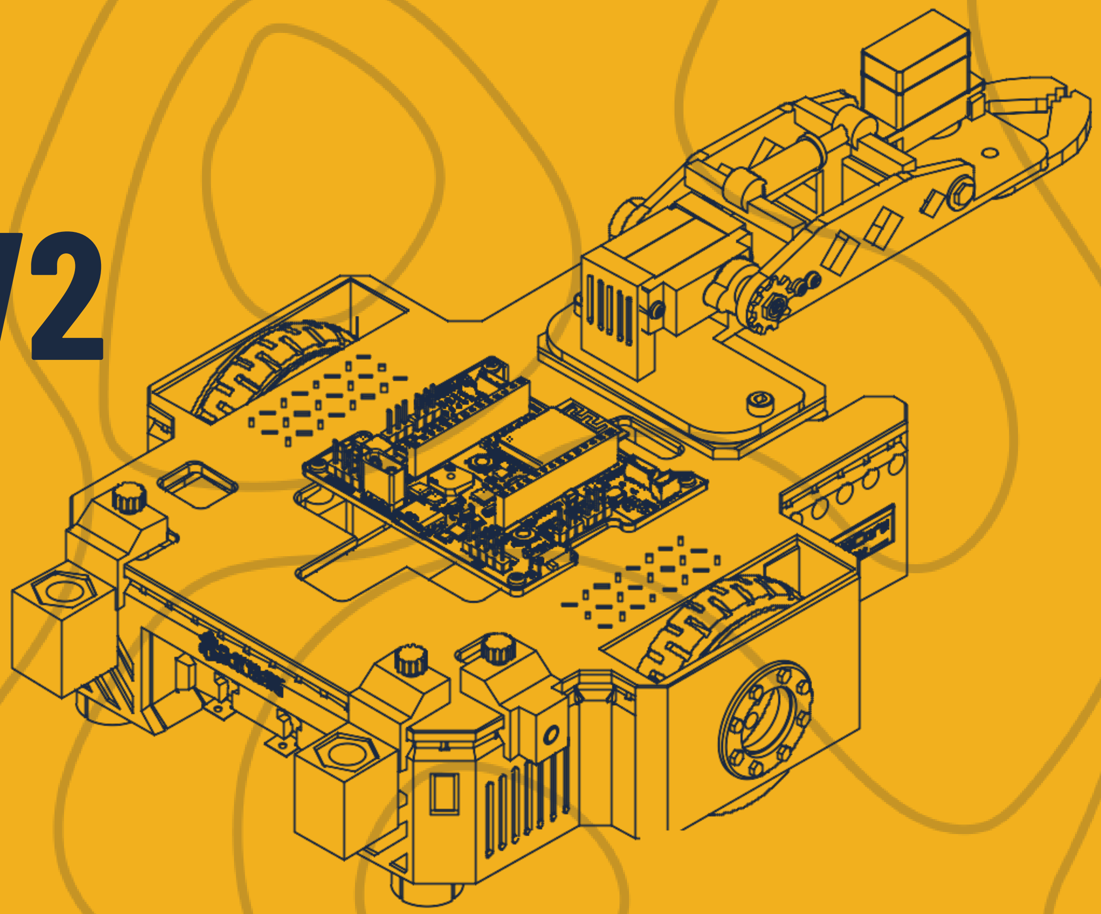
CLASSROOM ROBOT · REV 2.0 · DESIGNE

bCore
Zambia

BOCORE V2

OOM ROBOT

e. Create.



with Wi-Fi and Bluetooth

der motors, four servo ports, IMU

ltrasonic sensing

le with gear-linked claw

IN ZAMBIA

A · PLASTAL-BOT BUILDERS · LUSAKA

PLASTAL-BOT BUILDERS

Rob

Z

ROB

CLASSROOM

Build. Code. Cr

- ESP32-S3 with W
- Two encoder mo
- Line and ultrason
- Arm module with

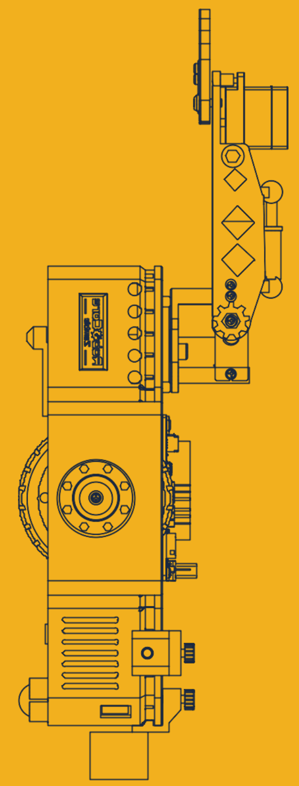
DESIGNED IN ZA

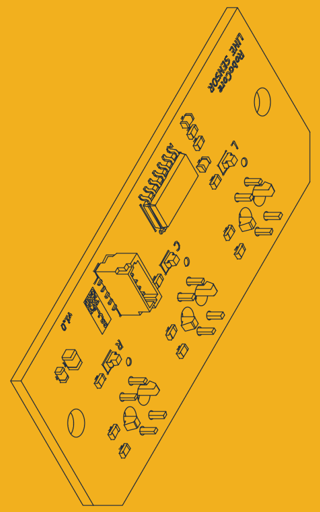
ROBOCORE ZAMBIA · PLA

E V2

SIGNED IN ZAMBIA

RoboCore V2 box, outside, tile 2 of 6 (row 1, column 2), 1:1. Overlap 10 mm with the neighbouring tiles.





PLASTAL-BOT BUILDERS

ROBOCORE V2

CLASSROOM ROBOT · REV 2.0

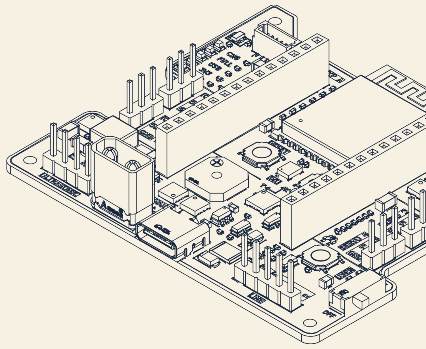
DESIGNED IN ZAMBIA

RC2.0 · 2026 · plastalbotbuilders.com

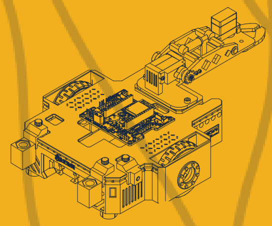
MEET ROBOCORE V2

A classroom robot designed and built in Zambia. The RoboCore V2 controller puts an ESP32-S3 with Wi-Fi and Bluetooth, two encoder motor drives, four servo ports, a motion sensor and sockets for line and ultrasonic sensing on one board, running from a single lithium cell with USB-C charging. The robot arrives assembled; the arm module bolts on in minutes.

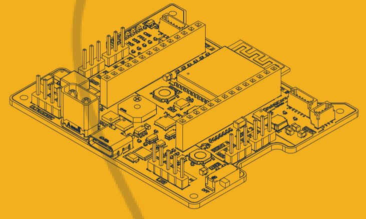
- ESP32-S3 with Wi-Fi and Bluetooth
- Two encoder motors, four servo ports, IMU
- Line and ultrasonic sensing
- Arm module with gear-linked claw



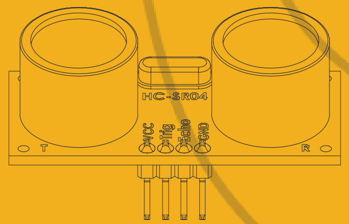
WHAT'S IN THE BOX



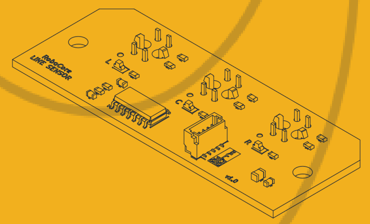
ROBOT, ASSEMBLED



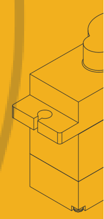
ROBOCORE V2



ULTRASONIC



LINE SENSOR



2 SERVO

RoboCore V2 robot, assembled, with controller, HC-SR04 ultrasonic, 3-channel line sensor and battery holder · Arm module: base, boom, claw, 2 micro servos, 2 ballast clips · USB-C cable · Screw
Quick-start card. 18650 cells and the M8 ballast nuts are not included.

SPECIFICATIONS

Controller RoboCore V2, ESP32-S3, 68 x 60 mm
Power 1S Li-ion, 3 x 18650 in parallel, USB-C charging
Drive 2 x TT encoder motors, 56 mm wheels, 2 casters

Sensing 3-channel line sensor, HC-SR04 ultrasonic, IMU
Robot 167 x 166 x 65 mm, about 540 g with the arm
Box 273 x 203 x 83 mm, E-flute card

DOCS & CODE

780126 RC2.0

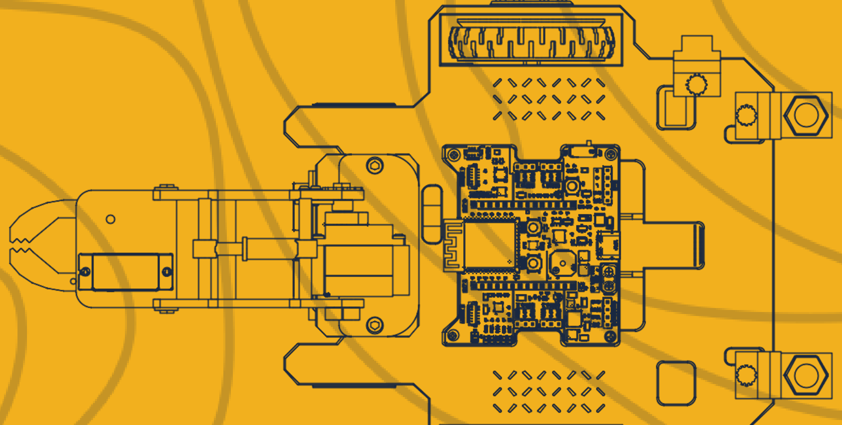
Not a toy. Contains small parts and screws: not for children under 8. Use only 1S lithium cells; charge through USB-C under supervision. Keep metal away from the antenna.

DESIGNED IN Z

RoboCore Zambia · Lusak

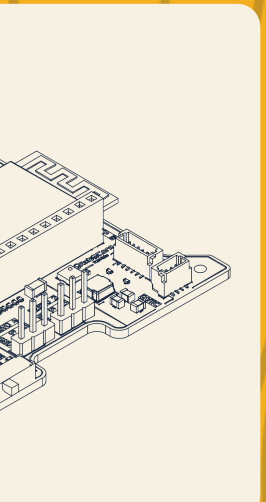
an build, program and repair

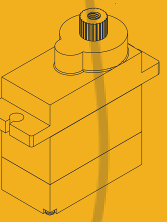
| CODE | CREATE |



RoboCore V2 box, outside, tile 3 of 6 (row 2, column 1), 1:1. Overlap 10 mm with the neighbouring tiles.

Core
bia





2 SERVOS + ARM

· Screw and nut pack ·

LASTAL-BOT
BUILDERS

SIGNED IN ZAMBIA

bia · Lusaka · 2026 · Rev 2.0

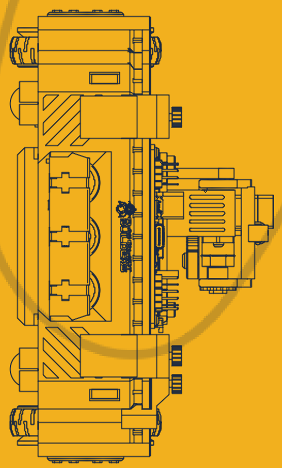
A robot a class can build

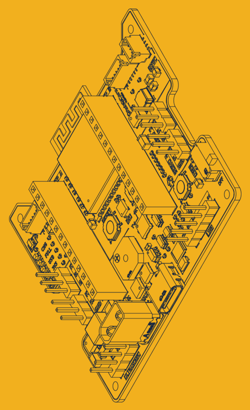
BUILD |

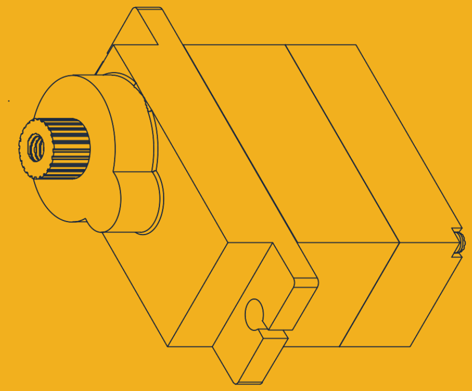
ROBOCORE V2

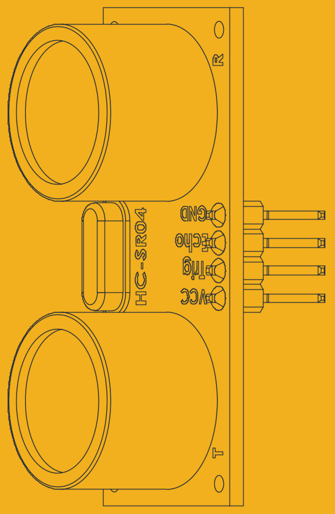
CLASSROOM ROBOT · REV 2.0

RoboCore
Zambia

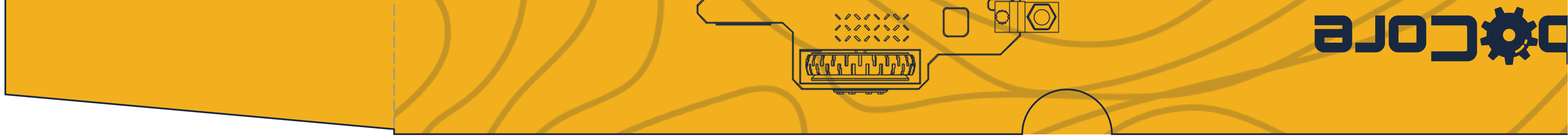








RoboCore V2 box, outside, tile 4 of 6 (row 2, column 2), 1:1. Overlap 10 mm with the neighbouring tiles.



RoboCore V2 box, outside, tile 5 of 6 (row 3, column 1), 1:1. Overlap 10 mm with the neighbouring tiles.

RoboCore V2 box, outside, tile 6 of 6 (row 3, column 2), 1:1. Overlap 10 mm with the neighbouring tiles.